Assignment six

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Yippe, new preamble.

> **Problem** (Munkres 30.5(a)). Show that every metrizable space with a countable dense subset has a countable basis.

> **Lemma.** *Every separable metric space is second-countable.*

Proof. Let (Y, d) be a separable metric space.

Meaning there exists $D \subseteq Y$, countable, for which

$$\overline{D} = Y.$$

Recall that $\overline{D}$ is the set of all points $x \in Y$ such that every open neighborhood of x intersects D:

$$U \cap D \neq \emptyset$$

for every open neighborhood U of x.

We have

$$D = \{x_1, x_2, \ldots\}.$$

Take

$$\mathscr{B} = \{B_q(x_i) \mid i = 1, 2, \ldots,\ q \in \mathbb{Q}_{>0}\}.$$

These are open sets and the set is countable since we are indexing over $\mathbb{N} \times \mathbb{Q}_{>0}$ which is countable.

It is not hard to see that this covers the space.

Suppose $x \in Y$. Fix $x_i \in D$ and choose $q \in \mathbb{Q}_{>0}$ such that

$$q > d(x, x_i).$$

Then

$$x \in B_q(x_i).$$

Now suppose V is an open set.

Then, for $x \in V$, there exists $\varepsilon > 0$ so that

$$B_\varepsilon(x) \subseteq V.$$

Since D is dense, there exists $x_i \in D$ such that

$$d(x, x_i) < \varepsilon/3.$$

Choose $r \in \mathbb{Q}_{>0}$ such that

$$d(x, x_i) < r < \varepsilon - d(x, x_i).$$

Such a rational exists since

$$2d(x, x_i) < \frac{2\varepsilon}{3} < \varepsilon.$$

Then $x \in B_r(x_i)$, and for every $y \in B_r(x_i)$ the triangle inequality gives

$$d(x, y) \leq d(x, x_i) + d(x_i, y) < d(x, x_i) + r < \varepsilon.$$

Thus

$$x \in B_r(x_i) \subseteq B_\varepsilon(x) \subseteq V.$$

So $\mathscr{B}$ is a countable basis for Y, making Y second-countable.

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Proof of 30.5(a). Suppose X is a metrizable space with a countable dense subset. That is,

$$X \cong Y$$

for some metric space (Y, d) and there exists $D \subseteq X$ such that

$$\overline{D} = X$$

with $|D| \leq \aleph_0$.

Let $\varphi : X \to Y$ denote the homeomorphism between them. Then

$$\begin{aligned} \varphi(\overline{D}) &= \overline{\varphi(D)} \\ \varphi(X) &= \overline{\varphi(D)} \\ Y &= \overline{\varphi(D)} \end{aligned}$$

so $\varphi(D)$ is dense in Y and, since φ is bijective, $|\varphi(D)| \leq \aleph_0$, so $\varphi(D)$ is a countable dense subset of Y.

Hence Y is separable.

By the lemma above, Y is second-countable.

Thus there exists a countable basis

$$\mathscr{B}_Y = \{B_1, B_2, \ldots\}$$

of Y.

Then, for any open set U in X, since $X \cong Y$, we have

$$U = \varphi^{-1}(V)$$

for open $V \subseteq Y$. But

$$V = \bigcup_{i \in I} B_i$$

for some $I \subseteq \mathbb{N}$ so

$$\begin{aligned} U &= \varphi^{-1}\left(\bigcup_{i=1}^{\infty} B_i\right) \\ &= \bigcup_{i=1}^{\infty} \varphi^{-1}(B_i). \end{aligned}$$

Thus

$$\mathscr{B} = \{\varphi^{-1}(B_1), \varphi^{-1}(B_2), \ldots\}$$

is a countable basis of X, hence X is second-countable.

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